

Two-Axis Robotic Arm

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Top-Level Architecture Block Diagram

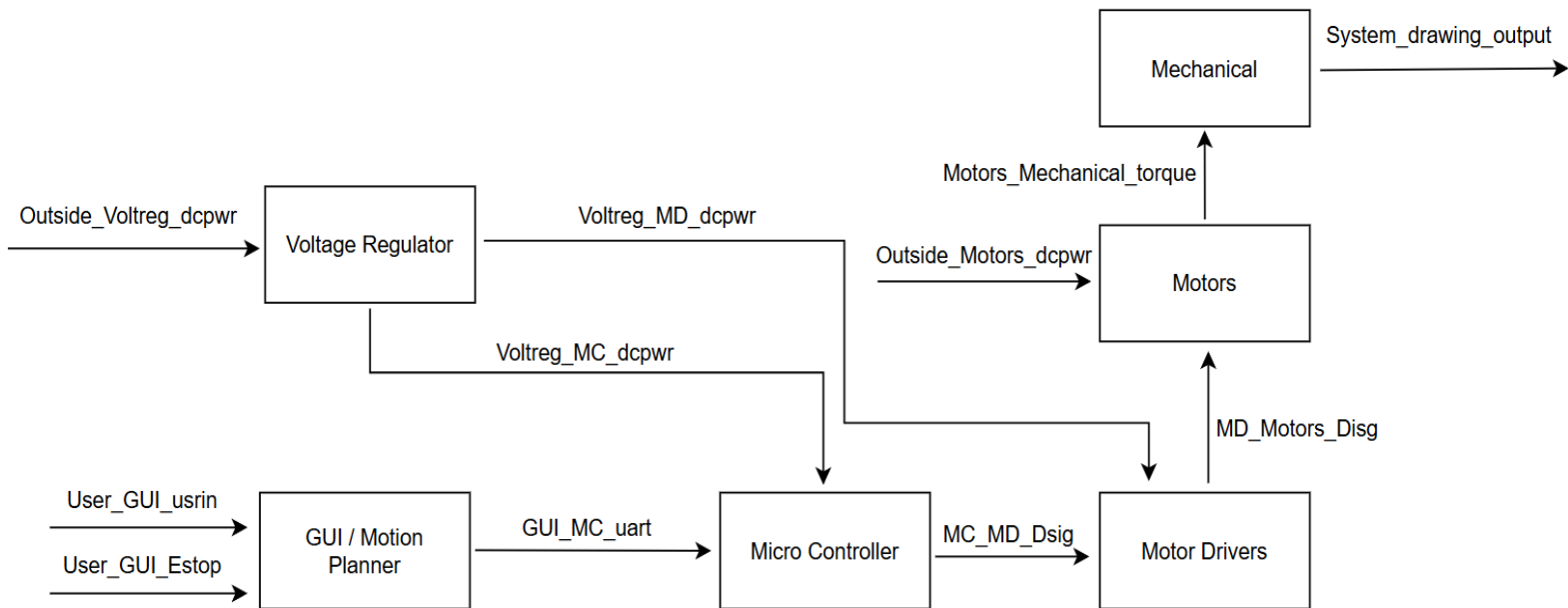


Fig 1. Top-Level Architecture Block Diagram.[1]

Block 1 Video Link

https://media.oregonstate.edu/media/t/1_diliv5mp

Block 1 Description

This voltage regulator is connected to an external adapter. The external adapter regulates the voltage from a 120V outlet to provide a fixed 12V output with a maximum current of 8A; the voltage regulator then converts this to a 5V output with a maximum current of 3A, which is supplied to the MCU. Additionally, the system is capable of delivering an output of up to 11.5V at 8A to the motor drivers used for motor control. Three motor drivers are utilized, each drawing a peak current of 2.8A.

To guard against reverse voltage, the power passes through a Schottky diode; this process results in a voltage drop, meaning that approximately 11.5V is the actual voltage delivered to the LM2596.

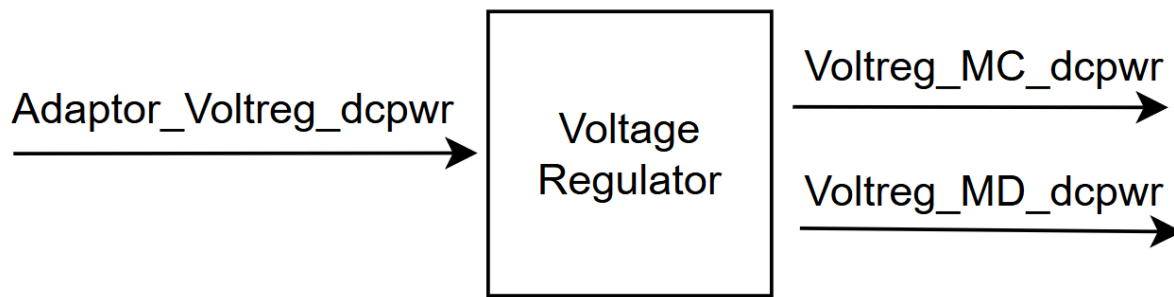


Fig. 2. Block-Level block diagram of Voltage Regulator.[1]

Block 1 Design Details

This voltage regulator block receives power from a 12V, 8A DC power adapter via a barrel connector. This block serves as the power management unit for the entire system. Input power is applied through the `Adaptor_Voltreg_dcpwr` interface; after passing through the barrel connector, it first proceeds through a Schottky diode and an electrolytic capacitor for input protection and filtering. The Schottky diode prevents reverse voltage [1], while the capacitor smooths out any ripple voltage that may originate from the external adapter. Following this stage—and accounting for the voltage drop across the Schottky diode—a stable 11.5V DC power supply is effectively established.

The resulting 11.5V power supply branches into two distinct paths within the system. The first path, designated `Voltreg_MD_dcpwr`, connects directly to three stepper motor drivers. Motor operation demands the highest current draw in the system (with a peak current of 2.8A per driver) [2]. To ensure a smooth and consistent current supply even under maximum load conditions, this path was designed to deliver power directly to the motors, bypassing any voltage regulators.

The second path feeds into an LM2596s switching buck converter to provide power to the microcontroller. Internally, this chip steps down the 11.5V input voltage to 5V. This stepped-down 5V output is then delivered as a stable DC power supply to the MCU via a catch diode, an inductor, and an output capacitor [3].

A key design feature of this system is the use of a variable-output voltage regulator—rather than a fixed 5V regulator—to generate an output voltage of 5.2V. I used the formula $V_{out} = V_{ref} * (1 + R2/R1)$. To ensure the resistors were as easily obtainable as possible, I utilized a 3.24kΩ SMD resistor and a 1kΩ SMD resistor found in the TekBots inventory. Consequently, the final output is $1.23V * (1 + 3.24k\Omega/1k\Omega) = 5.215V$.

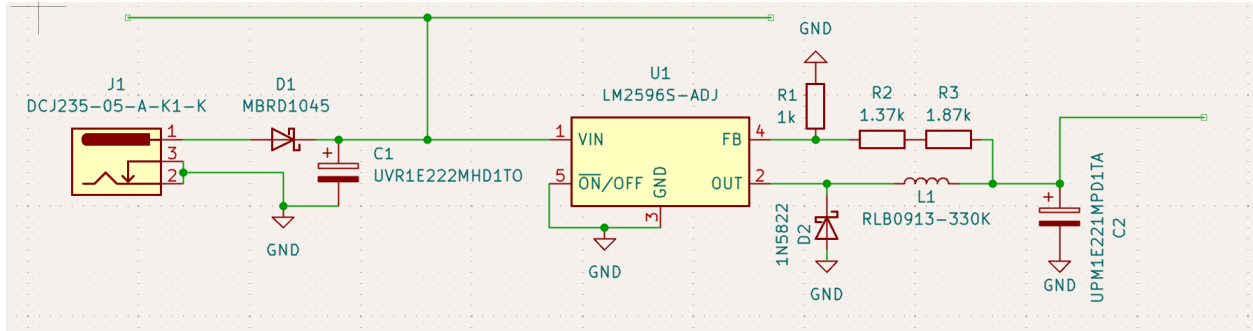


Fig 3. Schematic of Voltage Regulator.[1]

Block 1 Interface Validation

Interface property		
*Be sure to use the naming convention "from_to_type" here.	Why is this interface property this value?	How do you know your design details will meet or exceed this property? Cite your sources in IEEE.

Adaptor_Voltreg_dcpwr: input

Voltage: 12V	A voltage of 12V is required to smoothly drive the three NEMA 17 stepper motors, which serve as the system's primary power source.	The barrel jack connector and MBRD1045 Schottky diode are rated for voltages well exceeding 12V [1].
I _{peak} : 8.0A	Three stepper motors drawing up to 2.8A peak current each require a substantial current supply to prevent voltage drops under heavy load [2].	This is the instantaneous maximum current that the adapter must supply when the three motors accelerate simultaneously or are under maximum load [2].
I _{nom} : 6.0A	This is the sum of the average rated currents consumed by the three motors when the robotic arm performs typical movements.	The 8A adapter operates stably and without overheating issues in environments where it continuously supplies a rated current of 6A.

Voltreg_MD_dcpwr: Output

Vmax: 11.7V	This is the maximum voltage that can be applied to the driver, calculated by subtracting the minimum voltage drop across the diode from the adapter input voltage.	Since the maximum operating voltage of the TMC2209 driver is 28V, 11.7V falls well within a safe range [2].
Vmin: 11.0V	This is the minimum allowable limit for the voltage drop that may temporarily occur when the motor is rapidly driven.	A large-capacity capacitor of 2200uF stores and releases charge to prevent the voltage from dropping below 11V.
Inom: 6.0A	This is the rated current that flows continuously during motor operation.	The 8A adapter operates stably and without overheating issues in environments where it continuously supplies a rated current of 6A.

Voltreg_MC_dcpwr: Output

Vmax: 5.5V	This is the maximum voltage limit that the ESP32-DevKitC can withstand.	Even when accounting for the LM2596s and feedback resistor tolerances, the output exceeds 5.3 V [3] [4].
Ipeak: 3.0A	This represents the maximum current limit available for supply during MCU boot-up, during communication peaks, and to the driver logic power supply.	According to the LM2596-ADJ datasheet, the internal switching element guarantees an output current of up to 3A [3].
V_logic: 3.3V	The logic voltage level required for the ESP32 and the motor driver to communicate is 3.3V.	The internal regulator of the ESP32-DevKitC steps down the 5.2V power supply to 3.3V [4].

Block 1 Verification Process

1. Required Equipment

Digital Multimeter (DMM), 12V 8A Wall Adapter, Electronic Load

2. Setup and Input Interface Verification (Input: 12V 8A)

Setup:

1. Disconnect the provided PCB (Power Unit) from the MCU, motor driver, and motor.
2. Plug the 12V 8A wall adapter into a wall outlet and connect the DC barrel plug to the barrel connector on the PCB.
3. Set the DMM to DC voltage measurement mode.

Test:

1. Connect the DMM's red probe to the anode of the Schottky diode and the black probe to ground.
2. Verify on the DMM that the voltage falls within the acceptable range of V_{max} : 12.6V and V_{min} : 11.4V.
3. If the voltage falls within this range, the test passes; otherwise, it fails.

3. Setup and Input Interface Verification (Output: to Motor Driver)

Setup: Keep the adapter connected and prepare the electronic load equipment.

Test:

1. Connect the DMM's red probe to the positive terminal of the 2200 μ F capacitor. Connect the black probe to ground.
2. Verify that the voltage at the motor output terminal—after passing through the diode and undergoing a voltage drop—is between 11.7V and 11V.
3. Connect the electronic load between the positive terminal of the capacitor and ground. Set the load to 6A. Then, use the DMM to measure the voltage at that node again.
4. If the voltage remains at 11V or higher under the 6A load condition, the test succeeds; otherwise, it fails.

4. Configuration and Input Interface Verification (Output: to Microcontroller)

Setup: Disconnect the electronic load from the motor output terminal.

Test:

1. Connect the red probe of the DMM to the output terminal of the LM2596s and the black probe to ground.
2. Use the DMM to verify that the voltage remains between 5.1V and 5.3V.
3. If the voltage remains within this range, the test passes; otherwise, it fails.

Block 1 Artifacts

The peak current of a NEMA 17 motor is 2.8A, and we decided to incorporate three such motors into our design. Consequently, the instantaneous maximum current would reach $2.8A \times 3 + \alpha$. To address this, we had to discard the existing voltage regulator and design a new one from scratch. Furthermore, to withstand the high voltage and current levels involved, we were required to utilize a wide variety of numerous components.

During this process, I discovered that the TekBot kit did not contain the specific components I needed. As a result, the task of individually checking datasheets on Digi-Key to select the necessary parts proved to be quite arduous. For instance, designing the 5.2V supply line for the microcontroller required a 3.24k Ω resistor; however, a 3.24k Ω resistor was not available in the TekBot kit. To overcome this, I adopted a workaround by connecting a 1.37k Ω resistor and an 1.87k Ω resistor in series.

Through this experience, I was able to gain firsthand insight into the practical impact that component availability has on real-world engineering projects. Additionally, since the LM2596S chip would be unable to withstand the high 8A current output from the wall adapter, we had to design a separate, bypassed power supply route specifically for the motors.

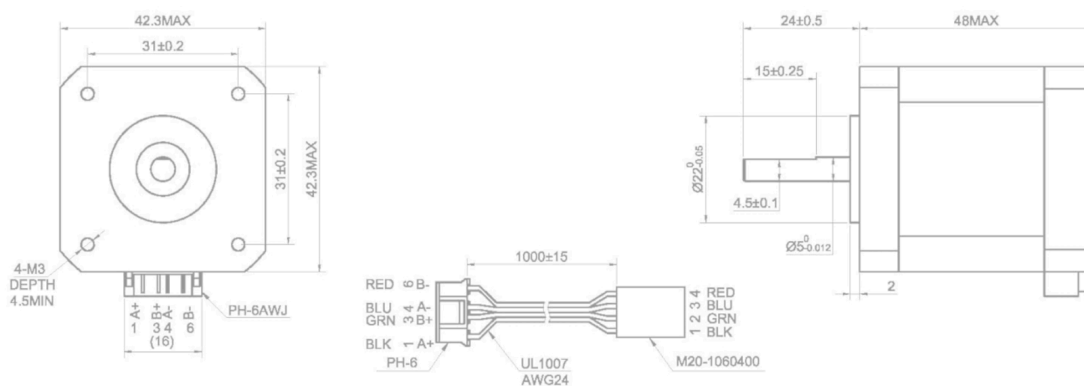


Fig 4. Nema 17 Stepper Motor.[1]

Created by Yunseong Cho

Part	Value	Footprint	Supplier
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Barrel Jack	DCJ235-05-A-K1-K	Connector_BarrelJack:BarrelJack_Kycon_KLDX-020 2-xC_Horizontal	Link
Schottky Diode	MBRD1045	Package_TO_SOT_SMD:TO-252-2	Link
Electrolytic Capacitors	UVR1E222M HD1TO	Capacitor_THT:CP_Radial_D12.5mm_P5.00mm	Link
Voltage Regulator	LM2596S-AD J	Package_TO_SOT_SMD:TO-263-5_TabPin3	Link
Catch Diode	1N5822	Diode_THT:D_DO-201AD_P15.24mm_Horizontal	Link
Output Capacitor	UPM1E221M PD1TA	Capacitor_THT:CP_Radial_D8.0mm_P3.50mm	Link
Output Inductor	RLB0913-33 0K	L_Radial_D8.7mm_P5.00mm	Link
R1	1k	Resistor_SMD:R_0603_1608Metric	Link
R2	1.37k	Resistor_SMD:R_0805_2012Metric	Link
R3	1.87k	Resistor_SMD:R_0805_2012Metric	Link
Female Header (esp32)	Conn_01x19	Connector_PinSocket_2.54mm:PinSocket_1x19_P2.54mm_Vertical	Link
Female Header (Motor Driver)	Conn_01x08	Connector_PinSocket_2.54mm:PinSocket_1x08_P2.54mm_Vertical	Link
Power Supply	12V, 8A		Link
Motor Driver	TMC2209 V1.3	4.75V - 28V, 2A(peak 2.8A)	Link
Motor 1	Nema17	12V, 2A	Link

Motor 2, 3	Nema17 Pancake	12V, 1A	Link
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Fig 5. Value and Footprint table.[1]

Block 1 Future Recommendations

Unlike the voltage regulator design project of the previous semester—for which specific guidelines were provided—this semester required us to start from a completely blank slate. The process of translating basic system requirements into a physically functional and verifiable circuit schematic is extremely challenging; in fact, it remains an ongoing endeavor that I have yet to fully complete.

Nevertheless, I experienced a profound sense of accomplishment after successfully implementing a 5.2V headroom. Through this process, I was able to cultivate a crucial engineering mindset: the ability to anticipate potential edge cases that might arise in real-world applications and to build in sufficient design margins to mitigate them.

If I could offer a piece of advice to my past self—specifically, to the version of me taking the JD1 course last semester—I would tell him this: before diving into the mathematical calculations and design work for a circuit, always verify first whether the necessary components are actually available and compatible with the other parts in the system.

Block 1 References

[1] ON Semiconductor, "MBRD1045: Schottky Power Rectifier, Surface Mount, 10 A, 45 V," MBRD1045 Datasheet, Oct. 2016.

[2] Trinamic, "TMC2209 Datasheet: Ultra-silent stepper motor driver with StallGuard4," Trinamic Motion Control GmbH, 2019.

[3] Texas Instruments, "LM2596 SIMPLE SWITCHER® Power Converter 150-kHz 3-A Step-Down Voltage Regulator," LM2596 Datasheet, May 2016.

[4] "ESP32 esp-dev-kits Documentation Release master Espressif Systems," 2025. Available: <https://docs.espressif.com/projects/esp-dev-kits/en/latest/esp32/esp-dev-kits-en-master-esp32.pdf>

Block 2 Video Link

{Create a record of your block working even if you plan to retest and improve the design. Update this link with the most-recent working verification process. The expectation is you iterate on your designs, but keeping a record will ensure you can show *something* to the instructional team even if everything falls apart at the last minute. This will be your backup verification if

something goes wrong or breaks between design, build, and final block testing. Make sure the instructional team has access to view the video.}

Block 2 Description

{Create a block-level block diagram of your individual block. Use IEEE figure labels. Add 1-2 paragraphs describing what this block does. What is its role in the system? How does its role relate to the overall system requirements? What is coming into the block? What does the block do to that input? What is created and then delivered as an output? How? This is where your deep dive into functionality goes. Make sure to include the names and functions of all interfaces related to this block and that they match your top-level architecture above.}

Block 2 Design Details

{Write a 1-2 page detailed description of how this block works. Include at least one schematic or other representative figure. Use IEEE figure labels. Demonstrate your learning by explaining clearly what the inputs are and where they come from. Explain how those inputs become outputs through the functionality of your block. Design details must include in-text citations in IEEE format. Cite resources from prior coursework, module resources lists from this class, or resources you have found externally.}

Block 2 Interface Validation

Interface property *Be sure to use the naming convention "from_to_type" here.	Why is this interface property this value?	How do you know your design details will meet or exceed this property? Cite your sources in IEEE.
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interface_name: input/output

Property 1	{Fill in each section of this table. Include citations, simulation results, calculations, and validation materials in each column that address the column prompt above.}	...
Property 2	...	...
Property 3	{Make sure you have a minimum of three properties per interface.}	...

interface_name: input/output

Property 1	{Fill in each section of this table. Include citations, simulation results, calculations, and validation materials in each column that address the column prompt above.}	...
Property 2	...	...
Property 3	{Make sure you have a minimum of three properties per interface.}	...

Block 2 Verification Process

1. {Enumerate a verification process here that any junior in the class could follow.
2. Include Setup and Test sections.
3. Be as specific and expository as possible and demonstrate each interface property from the validation table. Explicitly mention when each interface property is verified. What are the pass/fail conditions?
4. Imagine this process will be handed to another team to complete who did not design your system.
5. Write instructions they could follow.}

Block 2 Artifacts

{Populate this section with the miscellaneous but important findings that got you to your final block design. This means anything you had to learn in order to make choices about your design details. This might be prior coursework, examples found online, reference schematics, pseudocode, previous or prior version block diagrams, etc. Think of this section as a repository of your progress on this block.

If you're wondering what to put in this section, consider what someone looking at your design document would want to see to understand how you got to your final design decisions. This is unique to each block and design engineer.}

Block 2 Future Recommendations

{You have completed the individual technical challenges in this course. Take some time to reflect on how far you have come. How was this different from the Block 1 design cycle? What resources and skills did you utilize to improve during this second design-cycle? What would you tell yourself at the beginning of the term given what you know now? How will you use what you learned in this round in future classes or internships?}

Block 2 References

- [1] Include all relevant IEEE citations here. Be sure to cite them within the text. To get help with this you can search for “IEEE in-text citations” to understand how to cite in-text.
- [2] Cite everything you did not create yourself for this document. This includes but is not limited to diagrams, schematics, pseudocode/code, pinout visuals, etc.